FIITJEE (JEE-Advanced)

BATCHES – Two Year CRP-224 (All)

PHYSICS, CHEMISTRY & MATHEMATICS

QP Code: PET – 10

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. In Each Section is **One Part:** Part-A.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Sections.

(i) **Part-A (01-15)** contains (15) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking.**

Name of the Candidate :__________________________________

Batch :__________________ **Date of Examination :**__________________

Enrolment Number :__________________________________

SECTION-1 : PHYSICS

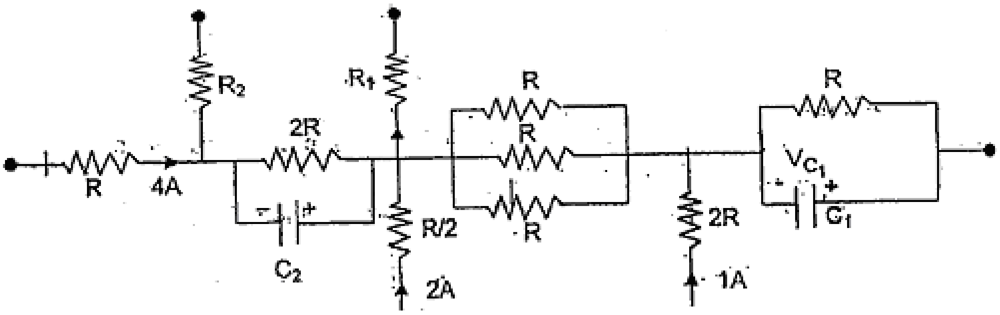
PART – A

(Numerical based)

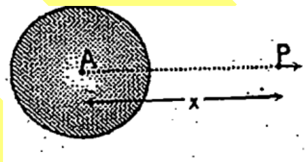
This section contains **15 questions**, numerical based questions, (answer of which maybe positive or negative numbers or decimals).

1. A soap bubble of radius R, surface tension T exists in equilibrium in zero atmospheric pressure outside it. How much charge (in SI units) should be uniformly distributed on its surface so that its radius in new equilibrium condition will be 2R. [Take $\pi^2 TR^3\varepsilon_0 = 1/12$, Here ε_0 is the absolute permittivity of the vacuum, consider temperature remains constant]

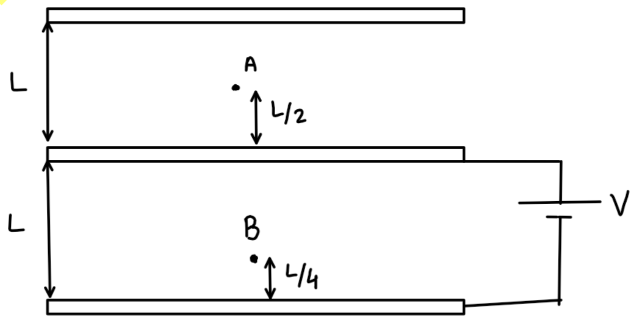
2. Here shown a diagram which is part of a network in steady state. Here R = 1 ohm. Potential difference across capacitor C_1 is V_{C_1}= 2 volts and Potential difference across capacitor C_2 is V_{C_2}= 4 volts. If currents in the resistance R_1 and R_2 are I_1 and I_2 respectively then find $\frac{I_1}{I_2}$.



3. A small dipole of dipole moment P is placed radially at a large distance x from the centre of sphere of volumetric charge density ρ and radius R as shown in the figure. The force acting on the dipole by the sphere is $\frac{k\rho R^3 P}{\varepsilon_0 x^3}$. Then find the value of k.

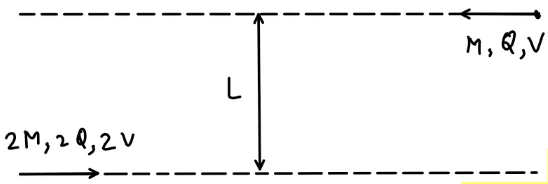


4. In a gravity free space, three identical metal plates have been placed parallel to each other as shown. In the initial state shown in the diagram the potential difference between the points A and B is V_1. Now, the middle plate is shifted down a distance L/2, keeping the battery connected. The other plates are kept fixed. The potential difference between the points A and B now becomes V_2. Find the value of V_1/V_2.



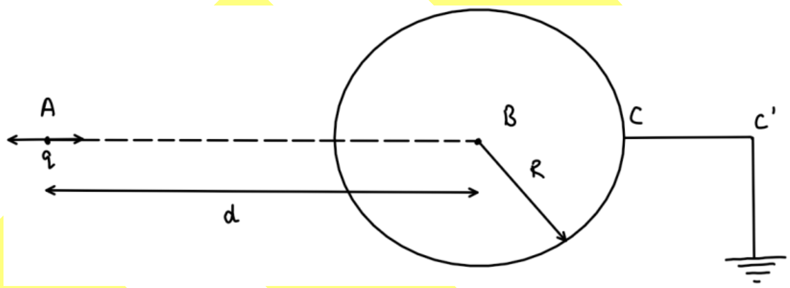
5. Consider a hypothetical condition in which charge is uniformly distributed in whole space. If net flux passing through the surface of the imaginary cone of base diameter D and slant length is Φ then if flux passing through the surface of imaginary sphere of radius D is KΦ. Then find K.

6. Two particle approach each other from very large distance as shown. (consider gravitational force between the particle is negligible) if $\frac{Q^2}{LMV^2} = 18\pi\varepsilon_0$ then the least separation between the particles is KL. Then find 'K'.



7. Six resistor, each of resistance R, are connected at their ends so as to form tetrahedral frame. Now two cells of negligible internal resistance and emf E is connected to the combination; One cell between any two corners and other cell between rest two corners. Find the thermal power developed in the circuit. (R = 8ohm, E = 3 Volts).

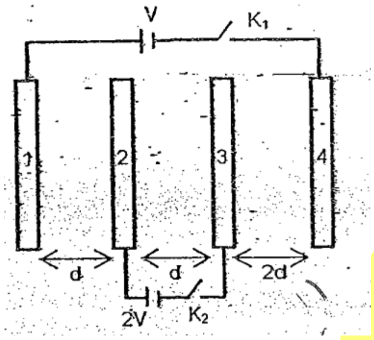
8. A charge q is undergoing SHM about point A along the line AB of a conducting sphere of radius R, amplitude of oscillation of charge a and angular frequency is ω. The sphere is grounded through wire CC'. Find the max value of current in the wire CC' in mA.
(take q = 2 Coulombs, ω = 5 rad/sec, R = 0.5m a = 1mm, d = 1m)



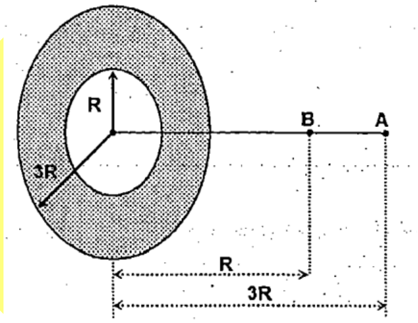
9. Consider a metallic sphere of radius R concentric with another hollow metallic sphere of inner radius 2R and outer radius 3R. The inner sphere is given a charge –Q and outer 2Q. The electrostatic potential energy of the system comes out to be $\frac{Q^2}{x\pi\varepsilon_0 R}$ then find x.

10. A charged particle having charge q and mass m is projected into a region of uniform electric field of strength E, Velocity V perpendicular to E. Throughout the motion apart from electric force particle also experiences a dissipative force of constant magnitude qE and directed opposite to its velocity. If magnitude of V is 5 m/s, then find its speed when it has turned through an angle of 90 degrees.

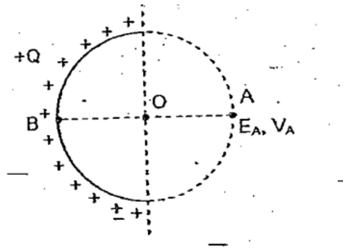
11. Four large metallic plates of area A each are kept parallel to each other with a small separation between them, as shown in the figure. A cell of emf V is connected across the two outermost plates through a switch K_1. The two inner plates are similarly connected with a cell of emf 2 V through a switch K_2. Initially both witches are open and the plates, starting from left to right (i. e. number 1 to 4) are given charges Q, 2Q, –2Q, –Q respectively. Find the charge appearing on plate 4 if both the switches are closed. (Q = 2 C, $\frac{\varepsilon_0 AV}{d} = Q$)



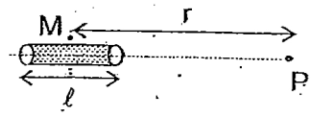
12. There is an annular disc of inner and outer radii R and 3R respectively. Charge is uniformly distributed with surface charge density σ on the surface of the annular disc. Then the minimum work done by an external agent to move a point charge from point A to point B on the axis of the disc is $\frac{kq\sigma R}{\varepsilon_0}$ then find 'k'.



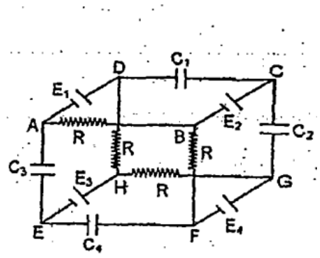
13. A non conducting wire is bent into a semi-circle of radius R and a charge +Q is uniformly distributed over it as shown in the figure. Find the ratio of magnitude of the Electric field at point A to the Potential at point A. (Take: R = 2 m)



14. A insulating cylindrical rod of diameter d and length l(l<<d) has a uniform surface charge density such that the electric field just outside the curved surface of the cylinder at point M is E. The electric field due to charge distribution at point P is $\frac{xEld}{r^2}$ (r>>l) then find the value of 'x'.



15. Four batteries of EMF E_1 = 4V, E_2 = 8V, E_3 = 12 V and E_4 = 16V, four capacitors with the same capacitance $C_1 = C_2 = C_3 = C_4$ = 1 μF, and four identical resistances are connected in circuit shown in figure. The internal resistances of batteries are negligible. At steady state find the total energy accumulated on the capacitors in μJ.

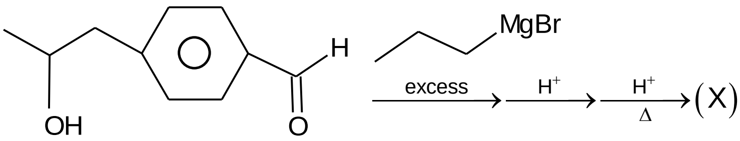


SECTION-2 : CHEMISTRY

PART – A

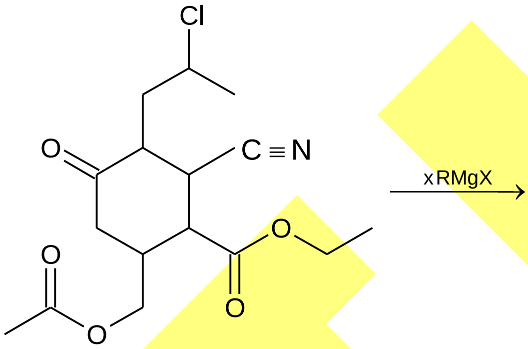
(Numerical based)

This section contains **15 questions**, numerical based questions, (answer of which maybe positive or negative numbers or decimals).

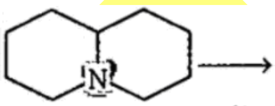
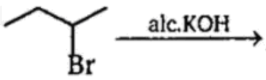
1. 

How many geometrical isomer of (X) is possible?

2. How many isomer of C_4H_8O when reacts with CH_3MgBr followed by acidification to give 2° alcohol(only consider carbonyl isomers)?
(including stereoisomer)

3. Total number of RMgX are consumed in the following reaction = x. Find x?


4. How many isomer of $C_4H_{10}O$ reacts with CH_3MgBr to evolve CH_4 gas? (Excluding stereoisomer)

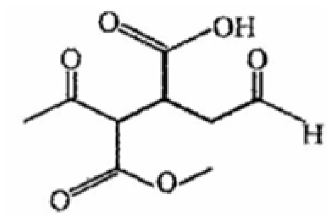
5.  x is total number of HEM (Hoffman Exhaustive Methylation and eliminations) to remove nitrogen from given compound.


y is total number of possible E_2 product (including stereoisomer)
Sum of $x+y=$?

6. $CH_3-\underset{\underset{Br}{|}}{CH}-\overset{\overset{CH_3}{|}}{CH}-CH_3 \xrightarrow{EtOH} (x)$ $(S_{N^1} + E_1)$ products. (including stereoisomer)
consider all products

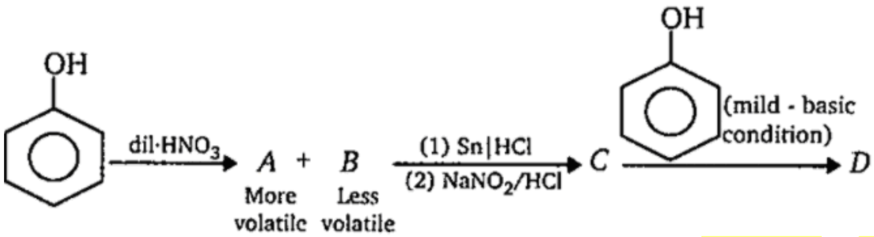
Total number of products are :

7.



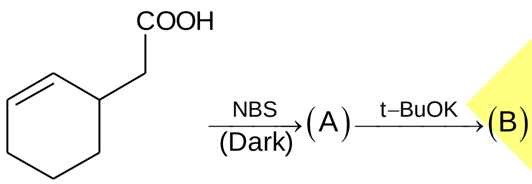
In the above given compound how many functional group reduced by LAH(Lithium aluminium hydride) and SBH(sodium borohydride) respectively. Find summation of functional groups reduced by LAH and SBH?

8.



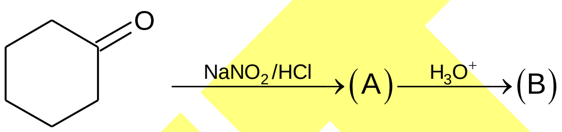
Double bond equivalent to D is

9.



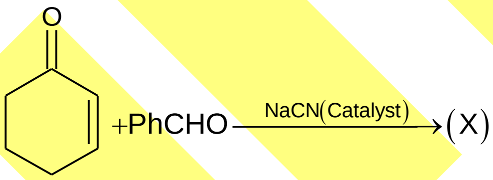
Summation of number of cycles in (A) + (B) = x. Find value of $\frac{X}{5}$

10.



If number of oxygen atom in (A) + (B) = X. Find value of $\frac{X}{10}$?

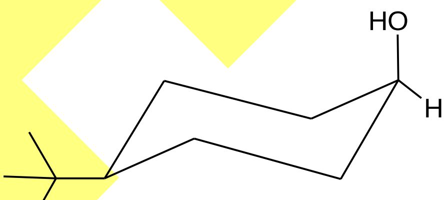
11.



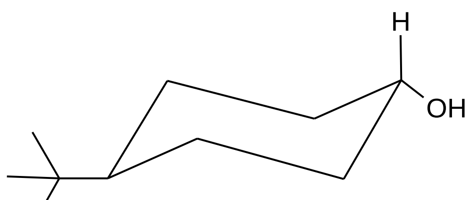
If degree of unsaturation = A

Find value of $\frac{A}{2}$

12.

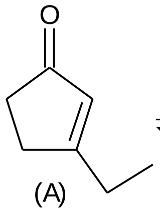


(1)



(2)

Which will undergo chromic acid oxidation easily.

13.  $\xrightleftharpoons{OH^-/H_2O} (B)$

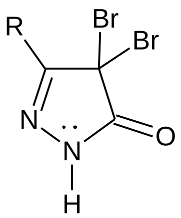
(A)

(A) and (B) are isomer of each other. If number of methyl group = X and no. of cycle = Y in compound (B). Find value of $\frac{X+Y}{2}$

14. $3PhCHO + 2\ddot{N}H_3 \xrightarrow{H^+(\text{Catalytic})} (A) + 3H_2O$

If number of pseudochiral carbon atoms = X

Find value of $\frac{X}{2}$

15.  $\xrightarrow[(2)\ H_3O^+]{(1)\ \ddot{\overline{O}}H/H_2O/\Delta} (A) + N_2 \uparrow$

If number of degree of unsaturation = X. In compound (A). Find value of $\frac{X}{4}$.

SECTION-3 : MATHEMATICS

PART – A

(Numerical based)

This section contains **15 questions**, numerical based questions, (answer of which maybe positive or negative numbers or decimals).

1. $\tan^{-1}\left[\frac{3\sin 2\alpha}{5+3\cos 2\alpha}\right]+\tan^{-1}\left[\frac{\tan\alpha}{4}\right]=\lambda\alpha$, then find the value of λ, where $-\frac{\pi}{2}<\alpha<\frac{\pi}{2}$.

2. Find number of solution of the equation $\left(\tan^{-1}x\right)^2+\left(\cot^{-1}x\right)^2=\frac{5\pi^2}{8}$.

3. Find number of integral solutions of the equation $[x][y]=x+y$. Here [.] denotes greatest integer function.

4. If domain of $f(x)=\frac{\sin^{-1}(\sin x)}{\sqrt{-\log_{\left(\frac{x+4}{2}\right)}\log_2\left(\frac{2x-1}{3+x}\right)}}$ is $(a,b)\cup(c,\infty)$, then find the value of $a+b+3c$.

5. Suppose $A_1, A_2, ..., A_{30}$ are thirty sets each having 5 elements and $B_1, B_2, ..., B_n$ are n sets each with 3 elements, let $\bigcup_{i=1}^{30} A_i=\bigcup_{i=1}^{n} B_j=S$ and each elements of S belongs to exactly 10 of the A'_is and exactly 9 of the B'_js. Then, n is equal to

6. If the sum $\sum_{n=1}^{10}\sum_{m=1}^{10}\tan^{-1}\left(\frac{m}{n}\right)=K\pi$, find the value of 'k'.

7. If $f(x)=x^3-3x+\sin^{-1}\left(a^2-3a+2\right)$. Then the smallest positive integer 'a' for which $f(x)=0$ has three distinct real solutions.

8. If the range of $f(x)=\left(\cos^{-1}\frac{x}{2}\right)^2+\pi.\sin^{-1}\frac{x}{2}-\left(\sin^{-1}\frac{x}{2}\right)^2+\frac{\pi^2}{12}\left(x^2+6x+8\right)$ is $\left[a\pi^2, b\pi^2\right]$, then find $2(a+b)$.

9. If $f:[2,\infty)\to[8,\infty)$ is a surjective function defined by $f(x)=x^2-(p-2)x+3p-2,\ p\in R$ then sum of values of p is $\left(m+\sqrt{n}\right)$ where $m,n\in N$. Find the value of $\frac{n}{m}$.

10. The polynomial $R(x)$ is the remainder upon dividing x^{2007} by x^2-5x+6. If $R(0)$ can be expressed as $ab\left(a^c-b^c\right)$. Find the value of $(a+b+c)$.

11. Find the number of functions that can be defined from set A = {1, 2, 3} to the set B = {1, 2, 3, 4, 5} such that $f(i) \le f(j) \forall i < j$

12. If the mean deviation of the numbers $1, 1+d, \ldots\ldots 1+100d$ from their mean is 255, then a value of d is:

13. If $x = \dfrac{4\ell}{1+\ell^2}$ and $y = \dfrac{2-2\ell^2}{1+\ell^2}$ where 'ℓ' is a parameter and range of $f(x,y) = x^2 - xy + y^2$ is: $[a, b]$; then $(a+b)$ is:

14. The period of the function $f(x) = \left(\sec^2\left(\dfrac{\pi x}{10}\right) - \tan^2\left(\dfrac{\pi x}{10}\right)\right)^{\cos^4 4\pi x + 100\{x\}}$ (where $\{.\}$ denotes fractional part function) is λ, then $\left(\dfrac{\lambda}{2}\right)$ is equal to:

15. Solution of the equation $\cot\left(\sum_{r=1}^{4} \cot^{-1} 2r^2\right) = \dfrac{3x+4}{3x+2}$ is equal to:

PERFORMANCE ENHANCEMENT TEST – 10

ANSWERS

SECTION-1 : PHYSICS

PART – A

1.	8	2.	0.50	3.	0.17 (Range: 0.16 – 0.17)		
4.	1.50	5.	18.48 (range: 18.45 – 18.50)				
6.	3.30 (range: 3.29 – 3.31)			7.	4.50	8.	1.25
9.	9.60	10.	2.50	11.	0.67 (range: 0.66 – 0.67)		
12.	0.34 (range: 0.32 – 0.35)			13.	0.25		
14.	0.25			15.	26		

SECTION – 2 : CHEMISTRY

PART – A

1.	4	2.	2	3.	7	4.	4
5.	6	6.	6	7.	6	8.	9
9.	0.8	10.	0.4	11.	3.5	12.	1
13.	1.5	14.	0.5	15.	0.75		

SECTION – 3 : MATHEMATICS

PART – A

1.	1	2.	1	3.	2	4.	5
5	45	6.	25	7.	1	8.	5
9.	2	10.	2011	11.	35	12.	10.1
13.	8	14.	5	15.	2		